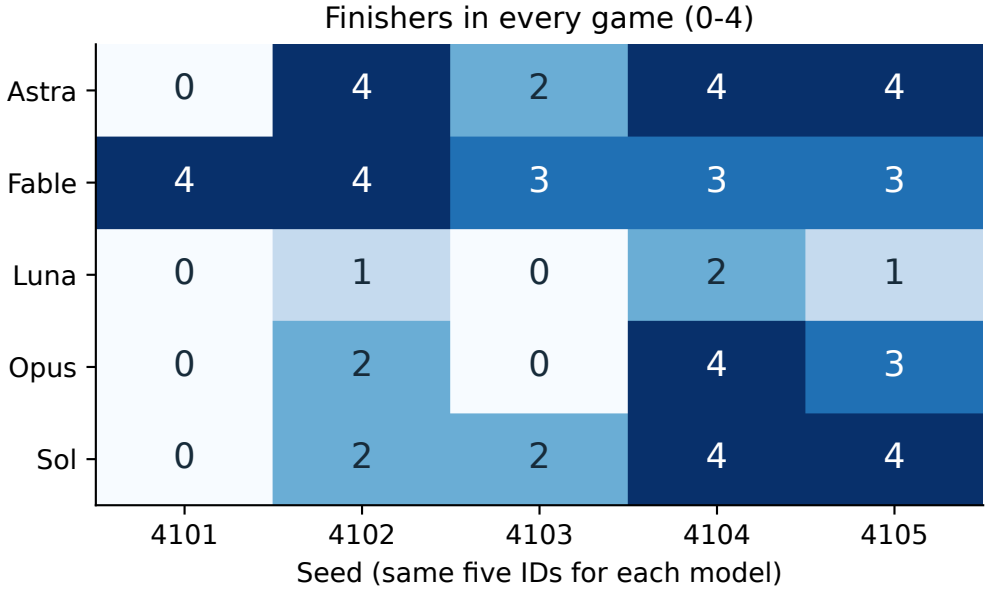
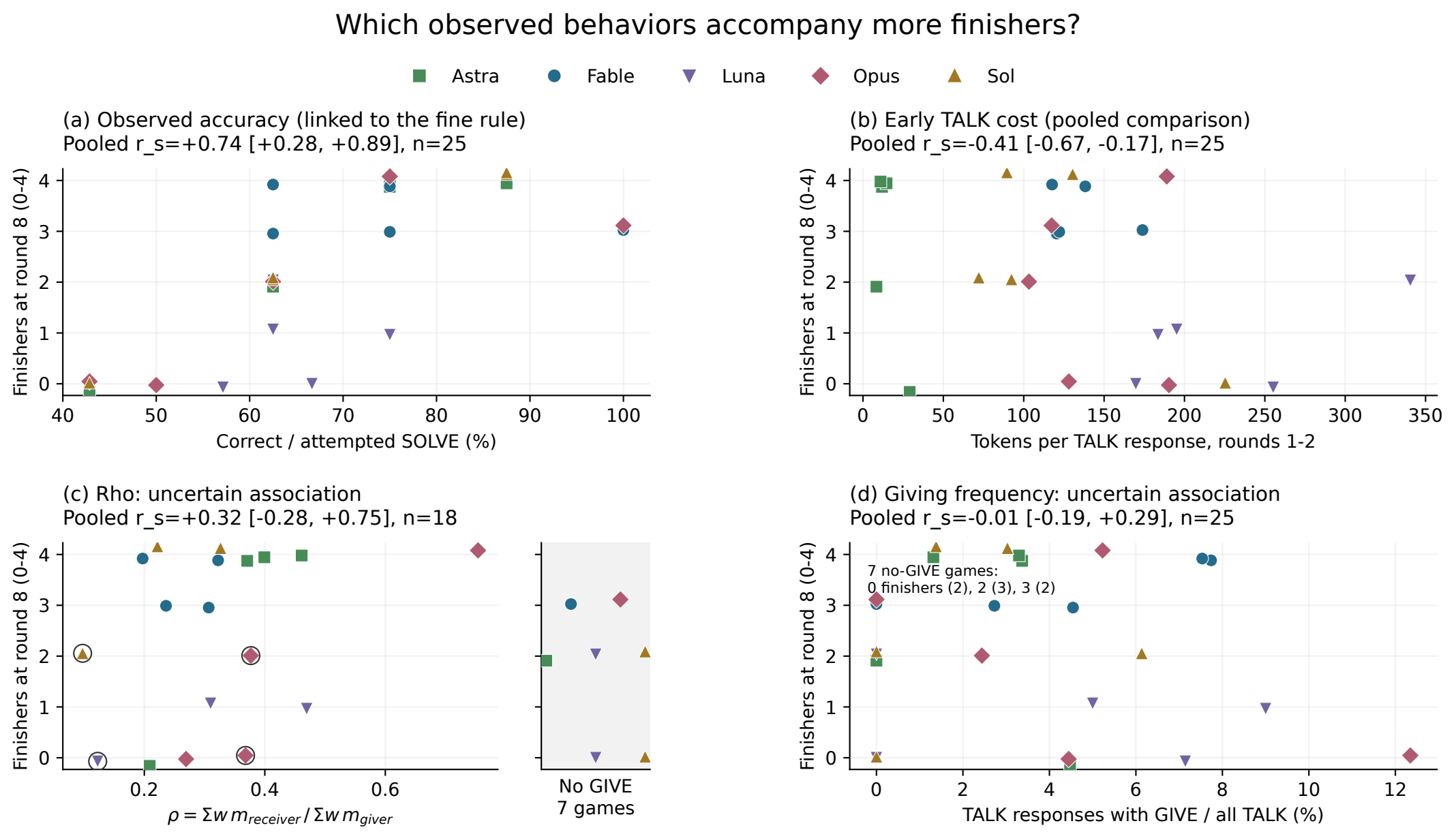
# Reaching the end is different from everyone reaching the end




| Model | >=1 finisher | All 4 finish | Finished | EXIT | Dead |
|---|---|---|---|---|---|
| Astra | 4/5 | 3/5 | 14/20 | 6 | 0 |
| Fable | 5/5 | 2/5 | 17/20 | 3 | 0 |
| Luna | 3/5 | 0/5 | 4/20 | 12 | 4 |
| Opus | 3/5 | 1/5 | 9/20 | 6 | 5 |
| Sol | 4/5 | 2/5 | 12/20 | 7 | 1 |

All 25 games: at least one finisher 19/25; all four 8/25. Seats: 56 finished + 34 EXIT + 10 dead = 100.

Completion means reaching the end of round 8, not eight correct answers. EXIT is alive but does not count as finishing.




Lower rho means the donor retains more relative margin in these data.

One point = one game; vertical offsets reveal overlaps. Non-finishers: 34 EXIT (alive) + 10 dead seats.

Brackets: 95% paired-seed bootstrap intervals, only 5 seed clusters. Pooled correlations mix within- and between-model patterns.

Accuracy is linked to outcomes by the fine rule, not a pure ability estimate (6-8 SOLVE attempts per game).

Early TALK uses rounds 1-2 to limit later survival selection (19-83 responses per game); it remains observational.

m = (post-GIVE balance - pending fine) / per-agent fine; w = GIVE amount. Negative individual margins are retained.

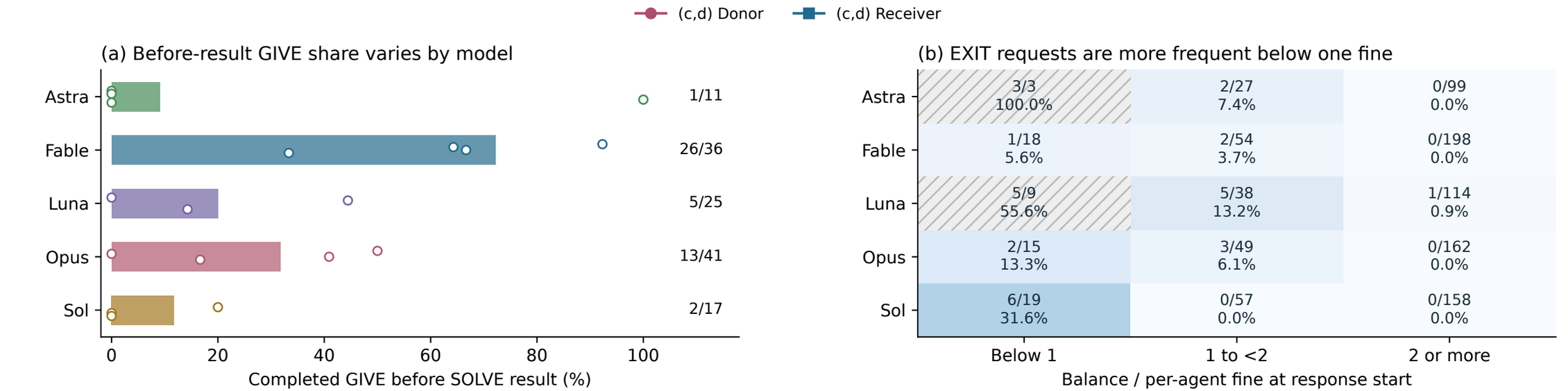
All 18 ratios have positive weighted numerator/denominator. Rings: 4 games contain negative individual margins.

Rho summarizes the game; it does not imply every transfer preserved the donor. No-GIVE is undefined, not zero.

Within-model TALK r_s: Astra -0.22, Fable -0.29, Luna +0.53, Opus -0.21, Sol -0.32 (5 games each).

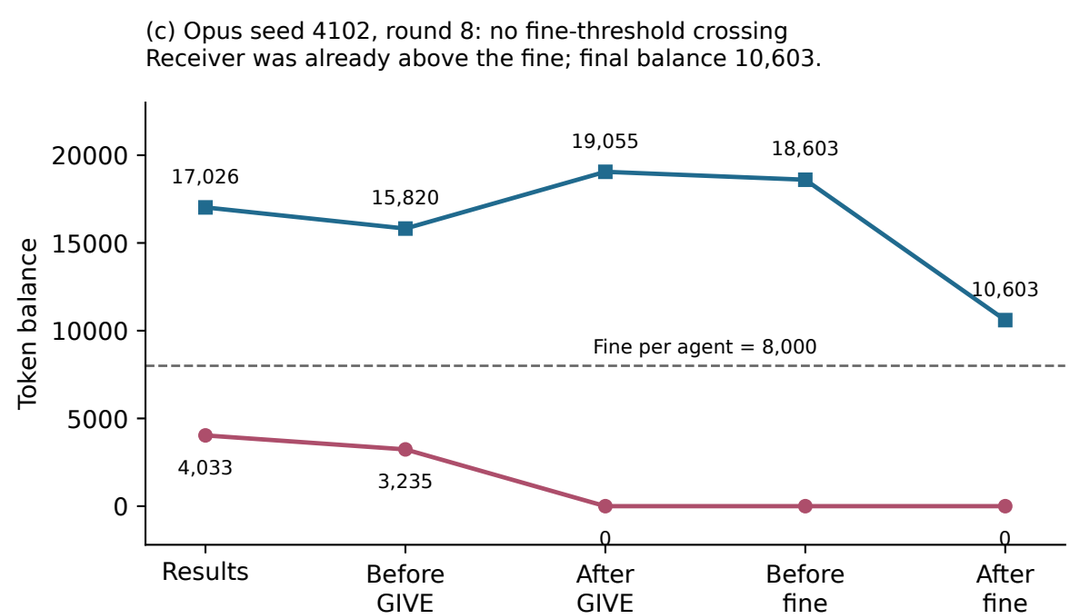
Giving frequency normalizes recorded response exposure; gross amount is shown in the companion figure.

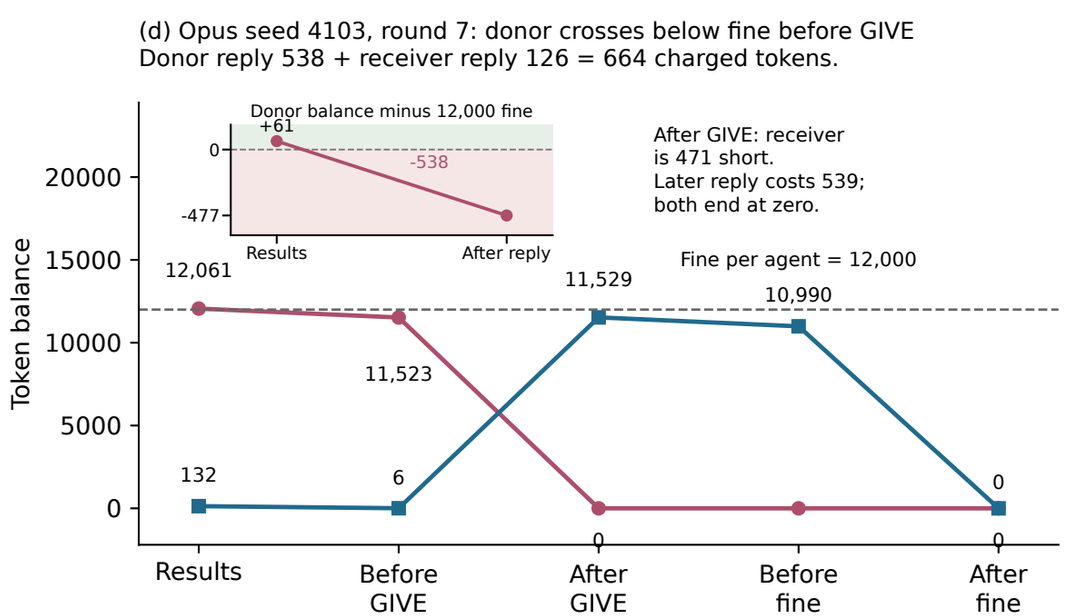
# Behavioral timing and two full-balance GIVE sequences




Bar + k/n: pooled events. Open points: games with GIVE.
No-GIVE games excluded here: Astra 1, Fable 1, Luna 2, Opus 1, Sol 2

Cells: accepted EXIT / non-solver prep replies (>=2 at round start).
Hatched grey: n < 10. Denominator counts repeated responses.
All models: 17/64 (26.6%), 12/225 (5.3%), 1/731 (0.1%).



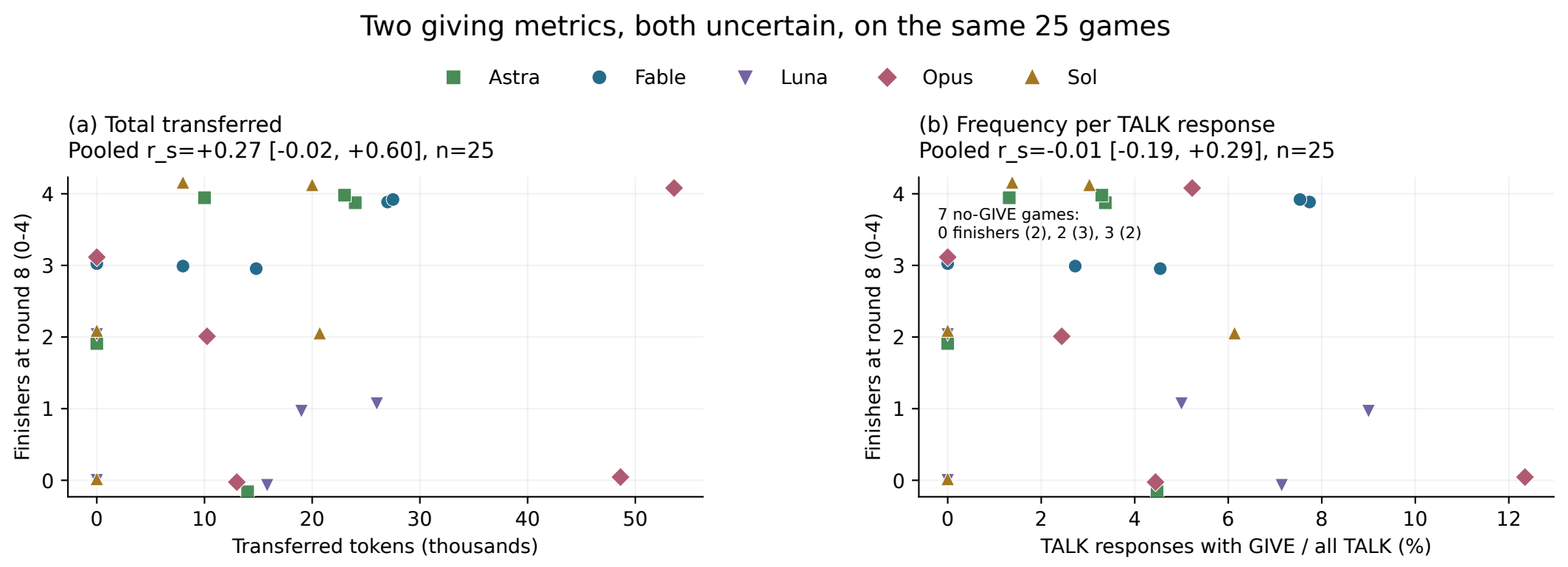

Cases (c,d) include every full-balance GIVE in the 130-transfer cohort (2/130); both belong to the same model, in different games.

GIVE changes neither agent's position above/below the pending fine in either case. These records do not show that giving caused death.

EXIT is voluntary departure while alive; it is unavailable after SOLVE. The case panels occur after SOLVE; death follows balance depletion.

Timing is descriptive, not proof of benefit. Fine in (b) is a potential round-start share; in (c,d) it is already due after an incorrect answer.

Panel (b) has 30 accepted requests; four lone-agent EXITs are excluded (one each in Astra, Luna, Opus, Sol). Total accepted EXITs = 34.




The same 25 games appear in both panels. Total amount counts recirculation; the response denominator is not a help-opportunity denominator.

Intervals use 5 paired seed clusters. Neither result identifies a causal benefit or absence of benefit from cooperation.